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B3.1 What happens during photosynthesis?

OCR B GCSE Biology (J257) Twenty First Century Science — Topic B3: Living together — food and ecosystems

1. The photosynthesis equation

Photosynthesis is the process by which plants and algae use light energy to convert carbon dioxide and water into glucose and oxygen. It takes place in the **chloroplasts**, which contain the green pigment **chlorophyll**.

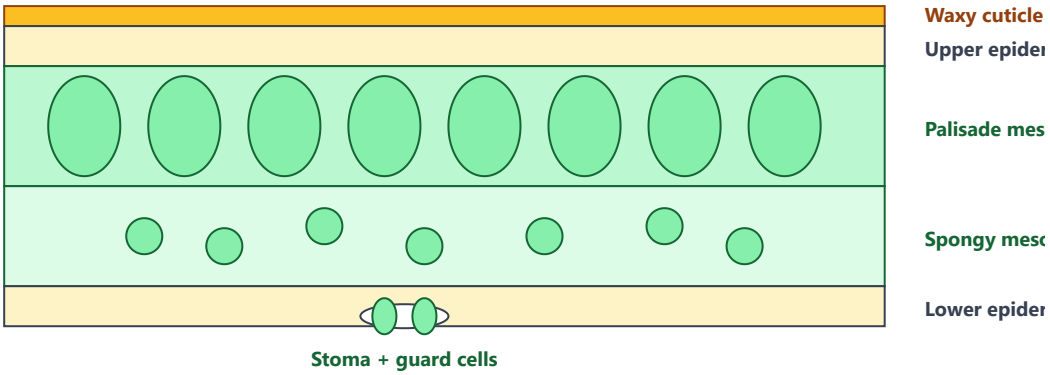
$$\text{Word: carbon dioxide} + \text{water} \rightarrow (\text{light}) \text{ glucose} + \text{oxygen}$$

$$\text{Symbol: } 6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$$

Photosynthesis is an **endothermic** reaction — it absorbs energy from light. The energy is stored in the chemical bonds of glucose.

2. Leaf structure and adaptations

Leaf cross-section



Adaptation	Function
Broad, flat shape	Maximum surface area for light absorption.
Thin	Short diffusion path for CO_2 and O_2.
Waxy cuticle	Reduces water loss.
Upper epidermis: transparent	Lets light through to the palisade cells.
Palisade mesophyll	Many chloroplasts, near top — most photosynthesis.
Spongy mesophyll: air spaces	Allow gas diffusion.
Stomata + guard cells	Open in light to allow CO_2 in / O_2 out; close in the dark or drought to save water.
Veins (xylem & phloem)	Xylem brings water; phloem carries glucose away.

3. Limiting factors

The rate of photosynthesis depends on three main factors. The factor in shortest supply at any moment is the **limiting factor**.

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Factor	Effect
Light intensity	Rate ↑ as light ↑ until another factor becomes limiting.
Carbon dioxide concentration	Rate ↑ as CO_2 ↑ until another factor limits (often around 0.4%).
Temperature	Rate ↑ with temperature up to optimum (~25-30 °C); enzymes denature above ~40 °C.
Chlorophyll concentration	Affected by mineral deficiency or disease (e.g. magnesium for chlorophyll).

Exam Tip: A graph that levels off shows another factor is limiting. To increase rate further, the new limiting factor must be raised. In commercial greenhouses, growers control all three factors (lights, heaters, CO_2 generators) — costs vs increased yield.

4. Inverse-square law (light)

$$\text{Light intensity} \propto 1 / d^2$$

Halving the distance from a lamp **quadruples** the light intensity. In photosynthesis practicals (PAG 5) plot rate against $1/d^2$ rather than against d.

5. Uses of glucose

Use	Detail
Respiration	Glucose broken down to release energy.
Stored as starch	Insoluble — does not affect water potential.

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Cellulose	Forms cell walls.
Lipids (oils)	Stored in seeds.
Amino acids → proteins	Combined with nitrate ions absorbed by roots.
Sucrose	Soluble form transported in phloem.

6. Summary checklist

- State the word and symbol equation for photosynthesis.
- Identify three limiting factors and explain how each affects rate.
- Describe four leaf adaptations.
- Apply the inverse-square law to a light-intensity calculation.
- List four uses of the glucose made in photosynthesis.

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